Model selection with neural networks and clustering data from simulations

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Motivation

In 2024, the DESI survey observations indicated a preference for dynamical dark energy models over standard ΛCDM [**adame2025desi**]. In the search for additional ways to validate or discard cosmological models using observational data, [**ocampo2025enhancing**] uses neural networks trained with mock growth rate data as likelihood-free model selection tool. We aim to expand on this analysis by including simulation measurements and a full-shape analysis of $P_m(k)$.

Full shape and growth rate

Changes in cosmology can noticeably affect the matter power spectrum. It can change the peak position, slope, or growth history. For that reason, we can approach the task of model selection using the full-shape at redshift $z = 0$ or the growth of structure.

For the growth history analysis, we can use the growth rate $f\sigma_8(z)$, where $f(z) \equiv d\ln\delta_+ / d\ln a$. It is common to fit this parameter as

$$f(z) = \Omega_m^{\gamma}(z), \tag{1}$$

where γ is sensitive to changes in the cosmological model and is ≈ 0.55 for ΛCDM

Modified Gravity and Simulations

The $f(R)$ self-accelerated gravity models replace the cosmological constant in Hilbert-Einstein action in the following way:

$$S = \int d^4x\sqrt{-g}\Big(\frac{R + f(R)}{2\kappa^2} + \mathcal{L}_m\Big). \tag{2}$$

The Hu-Sawicki gravity (HS) is particularly useful for clustering comparisons, since **it shares the same background history** as ΛCDM, and its deviation from it can be parameterized with a single constrainable value: f_{R0}

$$f(R) = -6\Omega_{\mathsf{DE},0}\frac{H_0^2}{c^2} + |\boldsymbol{f_{R0}}|\frac{\bar{R}_0^2}{R} + \dots \tag{3}$$

We use the **Quijote Simulations** [Quijote_sims] suite of N-body simulations for machine learning to obtain $P(k)$ measurements and their covariances for 2048 combinations of these parameters.

$\Omega_\mathsf{m} \in [0.10; 0.50]$	$\Omega_\mathsf{m} \in [0.10; 0.50]$
$\Omega_\mathsf{b} \in [0.03; 0.07]$	$\Omega_\mathsf{b} \in [0.02; 0.08]$
$h \in [0.50; 0.90]$	$h \in [0.50; 0.90]$
$n_s \in [0.80; 1.20]$	$n_s \in [0.80; 1.20]$
$\sigma_8(GR) \in [0.60; 1.00]$	$\sigma_8 \in [0.60; 1.00]$
$f_{R0} \in [-3 \times 10^{-4}; 0]$	

Table 1. Parameters for the HS simulations.

Table 2. Parameters for the ΛCDM simulations.

Building the growth rate

The growth factor δ_+ of perturbations is scale dependent in HS, so we take its value at $k_t = 0.1$ h/Mpc.

$$\delta_+(a) = \sqrt{P(k_t, a)/P(k_t, a = 1)}. \tag{4}$$

Then, to perform the logarithmic derivative, we fit it to 2nd degree polynomial, obtaining:

$$a\delta'_+(a) = \delta_+(a)\frac{d\ln\delta_+(a)}{d\ln a} = f\sigma_8(a)/\sigma_{8,0}. \tag{5}$$

Building The Neural Network

We use our growth rate and power spectra measurements to train a neural network for binary classification. The architecture of the model is a standard multi-layer perceptron, with three hidden layers and a 50% Dropout. We also included:

- **Optuna**: For hyperparameter tuning.
- **L1 regularization**: For generalization and contrast increase on the feature importance analysis.

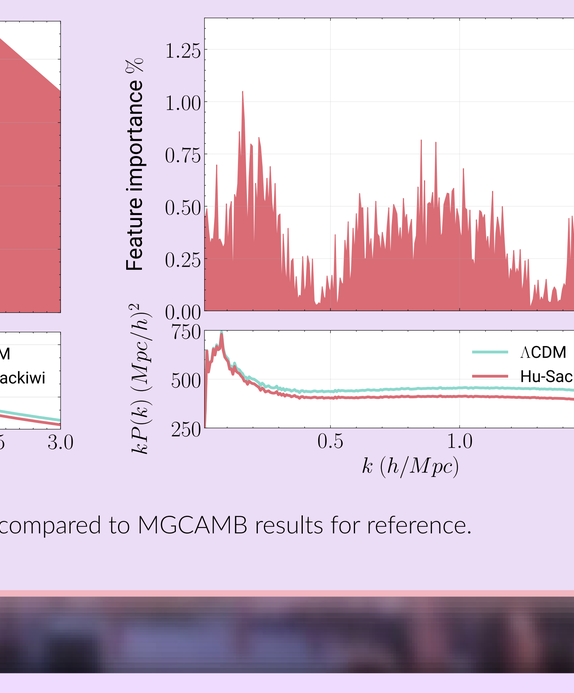


Figure 1. General architecture of the NN models used.

Neural Network Training and Performance

To train the neural networks, we add noise to the data using their respective covariance matrices. We obtain very accurate results from both approaches, with the full-shape model performing considerably better.

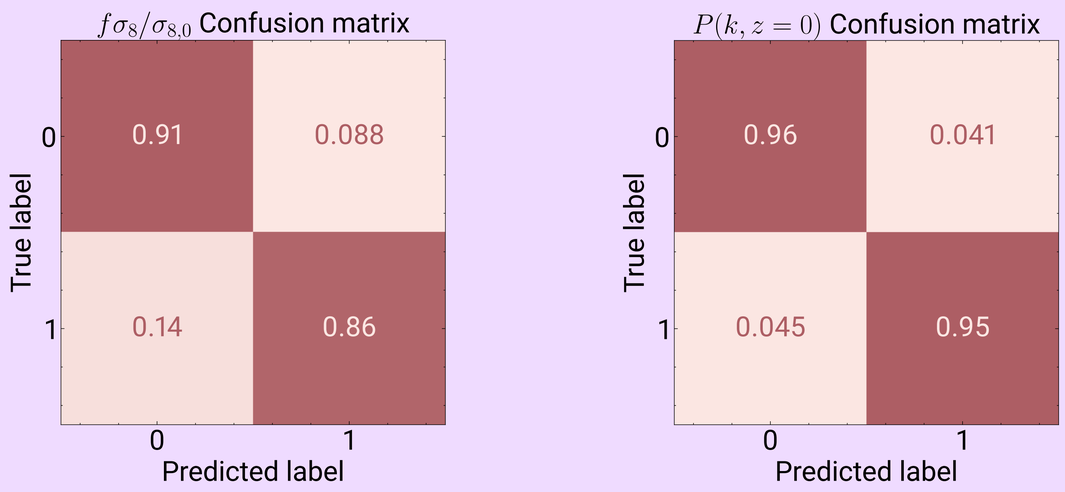


Figure 2. Confusion matrix for both models, where 0 is the label for ΛCDM and 1 for HS.

What is the model looking at?

Using the feature analysis tool Shap [**lundberg2017unified**], we find that:

- Not only low redshifts measurements are relevant for predictions based on growth rate.
- The models values the $P(k)$ derivatives at quasi non-linear and non-linear scales over the peak position

Figure 3. Feature analysis compared to MGCAMB results for reference.

Conclusions

- Neural networks are capable of accurately discern between models using the full shape $P(k)$ and its growth history.
- Feature analysis reflects some of the expected physical mechanisms present in the clustering measurements.

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References

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